

MATHEMATICAL ANALYSIS FOR COMPUTER SCIENCE

Mid Exam - Beginner Friendly Math Practice Note

CSE4105

Main terms English-e, baki explanation Banglish-e.

Important definitions easy English-e. Math: Formula $\rightarrow$ Method $\rightarrow$ Full Solve $\rightarrow$ Practice.

Ei PDF kivabe porba: First time porleo jeno bujha jay, tai prottek math-er age ki formula lagbe seta deya ache. Tarpor question dekhe kivabe method identify korba seta bola ache. Erpor ekdom step-by-step math. Finally same type practice deya ache. CT-er question je chapter-er, shei chapter-er moddhei solve kora.

Priority: *** = age korba; ** = tarpor; * = time thakle.

1. Recurrence Relation and Recursive Algorithm

1.1 Recurrence Relation

*** MUST DO

Definition (Easy English). A recurrence relation is an equation that expresses a problem, sequence, or running time in terms of smaller instances of the same problem.

Banglish bujha: Dhoro amra $T(n)$ directly jani na. Kintu $T(n-1)$ ba $T(n/2)$ jana thakle $T(n)$ ber korte pari. Ei relation-tai recurrence relation.

Base Case: Recurrence kothay stop korbe seta bole. Jemon $T(1) = 2$ ba $T(0) = c_0$. Base case chara repeatedly choto korte thakle final numeric/formula answer pawa jabe na.

1.2 CT Question: Linear Search recurrence derive kore complexity ber koro *** MUST DO

Formula / Rule First

$$T(n) = T(n-1) + c$$

Ekhane c holo ekta element check korar constant cost.

How to solve this type

Question-e jodi dekho prottek recursive call-e input n theke $n-1$ hocche, tahole recurrence repeatedly expand/unwind korba. $T(0)$ ba $T(1)$ pele stop korba. Tarpor n -er sathe total cost koto grow kore seta complexity.

Step 1: Recurrence derive kori.

Linear Search ekta element check kore. Eta constant work c . Element na pele baki $n-1$ element-e search kore.

$$T(n) = T(n-1) + c$$

Step 2: Ekbar expand kori.

$$T(n-1) = T(n-2) + c$$

Tai,

$$T(n) = T(n-2) + 2c$$

Step 3: Aro expand kori.

$$T(n) = T(n-3) + 3c$$

Pattern:

$$T(n) = T(n - k) + kc$$

Step 4: Base case-e stop kori.

Jodi base case $T(0) = c_0$ hoy, tahole

$$n - k = 0 \Rightarrow k = n$$

So,

$$T(n) = T(0) + nc = c_0 + nc$$

Constant ignore korle dominant part n .

$$\boxed{T(n) = O(n)}$$

Exam Writing A recurrence relation expresses the running time of a problem in terms of smaller inputs. For recursive Linear Search,

$$T(n) = T(n - 1) + c.$$

After k expansions, $T(n) = T(n - k) + kc$. Setting $n - k = 0$ gives $k = n$, so $T(n) = T(0) + nc = O(n)$.

Same-Type Practice

Solve:

$$T(n) = T(n - 1) + 5, \quad T(0) = 2.$$

Practice Solution

$$T(n) = T(n - k) + 5k.$$

Stop when $n - k = 0$, so $k = n$.

$$T(n) = T(0) + 5n = 2 + 5n.$$

$$\boxed{T(n) = 2 + 5n = O(n)}$$

1.3 CT Question: $T(n) = T(n/3) + 4$, $T(1) = 2$

***** MUST DO**

Formula / Rule First

$$T(n) = T(n/3) + 4, \quad T(1) = 2$$

How to solve this type

Ekhane prottek step-e n ke 3 diye divide kora hocche. Tai 1st expansion-e $n/3$, 2nd-e $n/3^2$, 3rd-e $n/3^3$. General step-e $n/3^k$. Base case $T(1)$ pawar jonno:

$$\frac{n}{3^k} = 1.$$

Step 1: Expand.

$$T(n) = T(n/3) + 4$$

Abar,

$$T(n/3) = T(n/9) + 4.$$

So,

$$T(n) = T(n/9) + 8 = T(n/3^2) + 2(4).$$

Step 2: Pattern likhi.

$$T(n) = T(n/3^k) + 4k.$$

Step 3: Base case condition.

$$\frac{n}{3^k} = 1$$

$$n = 3^k$$

$$k = \log_3 n.$$

Step 4: k bosai.

$$T(n) = T(1) + 4 \log_3 n$$

$$\boxed{T(n) = 2 + 4 \log_3 n}$$

Hence,

$$\boxed{T(n) = \Theta(\log n)}.$$

Same-Type Practice

$$T(n) = T(n/2) + 3, \quad T(1) = 1.$$

Practice Solution After k steps,

$$T(n) = T(n/2^k) + 3k.$$

Base case:

$$n/2^k = 1 \Rightarrow k = \log_2 n.$$

So,

$$\boxed{T(n) = 1 + 3 \log_2 n = \Theta(\log n)}.$$

1.4 Extra Important: $T(n) = T(n-1) + n$

**** IMPORTANT**

Formula / Rule First

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

How to solve this type

Ekhane prottek step-e same constant add hocche na. $n, (n-1), (n-2), \dots$ add hobe. Seshe summation formula use korba.

$$\begin{aligned} T(n) &= T(n-1) + n \\ &= T(n-2) + (n-1) + n \\ &= \cdots = T(0) + 1 + 2 + \cdots + n. \end{aligned}$$

Therefore,

$$T(n) = T(0) + \frac{n(n+1)}{2}.$$

Dominant term n^2 , so

$$\boxed{T(n) = \Theta(n^2)}.$$

Same-Type Practice

$$T(n) = T(n-1) + 2n, \quad T(0) = 0.$$

Practice Solution

$$T(n) = 2(1 + 2 + \cdots + n) = 2 \cdot \frac{n(n+1)}{2} = n(n+1).$$

$$\boxed{\Theta(n^2)}$$

1.5 Extra Important: Iteration Method $T(n) = c + T(n/2)$

**** IMPORTANT**

Formula / Rule First

$$T(n) = c + T(n/2).$$

How to solve this type

Input prottek step-e half hocche. Tai $n/2^k = 1$ condition use korba. Eta ager $n/3$ recurrence-er same pattern.

$$\begin{aligned} T(n) &= c + T(n/2) \\ &= c + c + T(n/4) \\ &= 3c + T(n/8) \end{aligned}$$

After k steps,

$$T(n) = kc + T(n/2^k).$$

Stop:

$$n/2^k = 1 \Rightarrow k = \log_2 n.$$

So,

$$\boxed{T(n) = c \log_2 n + T(1) = \Theta(\log n)}.$$

Same-Type Practice Solve $T(n) = 5 + T(n/2)$, $T(1) = 2$.

Practice Solution

$$T(n) = 5k + T(n/2^k), \quad k = \log_2 n.$$

$$\boxed{T(n) = 2 + 5 \log_2 n = \Theta(\log n)}.$$

2. Tower of Hanoi

2.1 Basic Idea and Rules

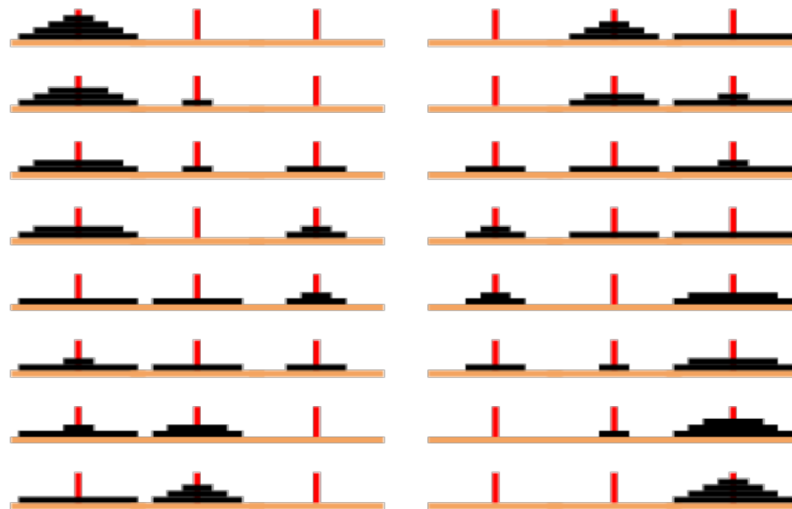
***** MUST DO**

Definition (Easy English). Tower of Hanoi is a puzzle with three pegs and n disks of different sizes. The goal is to move all disks from the source peg to the destination peg while following the movement rules.

Rules:

- Ekbar-e only one disk move kora jay.
- Shudhu top disk move kora jay.
- Larger disk ke smaller disk-er upor rakha jabe na.

How to solve the 4 disk



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Figure 1: *

Lecture slide visual: 4-disk Tower of Hanoi movement sequence. Image-ta dekhe disk movement order visualize korba; math-e minimum move count $2^4 - 1 = 15$.

2.2 CT Question: Recurrence derive kore $n = 4$ solve

*** MUST DO

Formula / Rule First

$$T(n) = 2T(n - 1) + 1, \quad T(1) = 1$$

Closed form:

$$T(n) = 2^n - 1.$$

How to solve this type

n disk move korte always 3 ta part:

1. Top $n - 1$ disk Source $\rightarrow$ Auxiliary.
2. Largest disk Source $\rightarrow$ Destination: exactly 1 move.
3. $n - 1$ disk Auxiliary $\rightarrow$ Destination.

Tai,

$$T(n) = T(n - 1) + 1 + T(n - 1) = 2T(n - 1) + 1.$$

Now $n = 4$:

$$T(1) = 1$$

$$T(2) = 2(1) + 1 = 3$$

$$T(3) = 2(3) + 1 = 7$$

$$T(4) = 2(7) + 1 = 15.$$

$$\boxed{\text{Minimum moves for 4 disks} = 15}$$

Shortcut: Direct closed form use korle:

$$T(4) = 2^4 - 1 = 16 - 1 = 15.$$

CT-er $n = 4$: Full 15-Move Sequence

Disk naming: Disk 1 = smallest, Disk 4 = largest. Source = A, Auxiliary = B, Destination = C.

Beginner idea: 4 disk-er jonno age top 3 disk A theke B-te nite hobe, tarpor Disk 4 A theke C-te, tarpor oi 3 disk B theke C-te. Nicher 15 ta move exam-e full movement sequence chaile use korte parba.

Move	Disk	From → To
1	1	A → B
2	2	A → C
3	1	B → C
4	3	A → B
5	1	C → A
6	2	C → B
7	1	A → B
8	4	A → C
9	1	B → C
10	2	B → A
11	1	C → A
12	3	B → C
13	1	A → B
14	2	A → C
15	1	B → C

Check: Total 15 moves, which matches

$$2^4 - 1 = 15.$$

Exam Writing For the CT answer, first write the three rules, then derive $T(n) = 2T(n-1) + 1$, then calculate $T(4) = 15$. If a movement illustration is required, draw/describe the 15 moves shown above.

Same-Type Practice

- (a) $n = 5$ hole minimum moves? (b) $n = 7$ hole minimum moves?

Practice Solution

$$T(5) = 2^5 - 1 = 31.$$

$$T(7) = 2^7 - 1 = 127.$$

2.3 Why $2^n - 1$?

**** IMPORTANT**

Small values:

$$1, 3, 7, 15, 31, \dots$$

Egula respectively

$$2^1 - 1, 2^2 - 1, 2^3 - 1, 2^4 - 1, 2^5 - 1.$$

Tai closed form pattern:

$$\boxed{T(n) = 2^n - 1}.$$

3. Lines in the Plane

3.1 Problem and Recurrence

*** MUST DO

Definition (Easy English). The Lines in the Plane problem asks for the maximum number of regions formed by n straight lines when no two lines are parallel and no three lines meet at one point.

Formula / Rule First

$$R(n) = R(n-1) + n, \quad R(0) = 1.$$

Closed form:

$$R(n) = \frac{n(n+1)}{2} + 1.$$

Keno $+n$? n -th new line previous $n-1$ line ke different points-e intersect kore. Ete new line-ta n ta segment-e vag hoy. Prottek segment ekta new region create kore. Tai previous region-er sathe n add hoy.

Small cases:

$$R(0) = 1, \quad R(1) = 2, \quad R(2) = 4, \quad R(3) = 7.$$

3.2 CT Question: Prove $R(n) = \frac{n(n+1)}{2} + 1$

*** MUST DO

How to solve this type

Induction proof-e 3 ta part mone rakhba: **Base Case** $\rightarrow$ **Induction Hypothesis** $\rightarrow$ **Induction Step.**

Step 1: Base Case, $n = 1$.

$$R(1) = R(0) + 1 = 2.$$

Formula-r RHS:

$$\frac{1(1+1)}{2} + 1 = 2.$$

So base case true.

Step 2: Assume formula true for $n = k$.

$$R(k) = \frac{k(k+1)}{2} + 1.$$

Step 3: Prove for $n = k + 1$.

Recurrence:

$$R(k+1) = R(k) + (k+1).$$

Assumption bosai:

$$\begin{aligned} R(k+1) &= \frac{k(k+1)}{2} + 1 + (k+1). \\ &= \frac{k(k+1) + 2(k+1)}{2} + 1 \\ &= \frac{(k+1)(k+2)}{2} + 1. \end{aligned}$$

Eta exactly formula-r $n = k + 1$ version. Therefore,

$$R(n) = \frac{n(n+1)}{2} + 1$$

is proved by mathematical induction.

Same-Type Practice

Maximum region for 6 lines and 8 lines ber koro.

Practice Solution

$$R(6) = \frac{6 \cdot 7}{2} + 1 = 22.$$

$$R(8) = \frac{8 \cdot 9}{2} + 1 = 37.$$

4. Josephus Problem

4.1 Definition and $n = 10$ CT Question

*** MUST DO

Definition (Easy English). In the Josephus problem, n people stand in a circle and every second person is eliminated repeatedly until only one survivor remains. $J(n)$ denotes the survivor's position.

For $n = 10$, elimination order:

2, 4, 6, 8, 10, 3, 7, 1, 9.

Only 5 remains.

$$J(10) = 5.$$

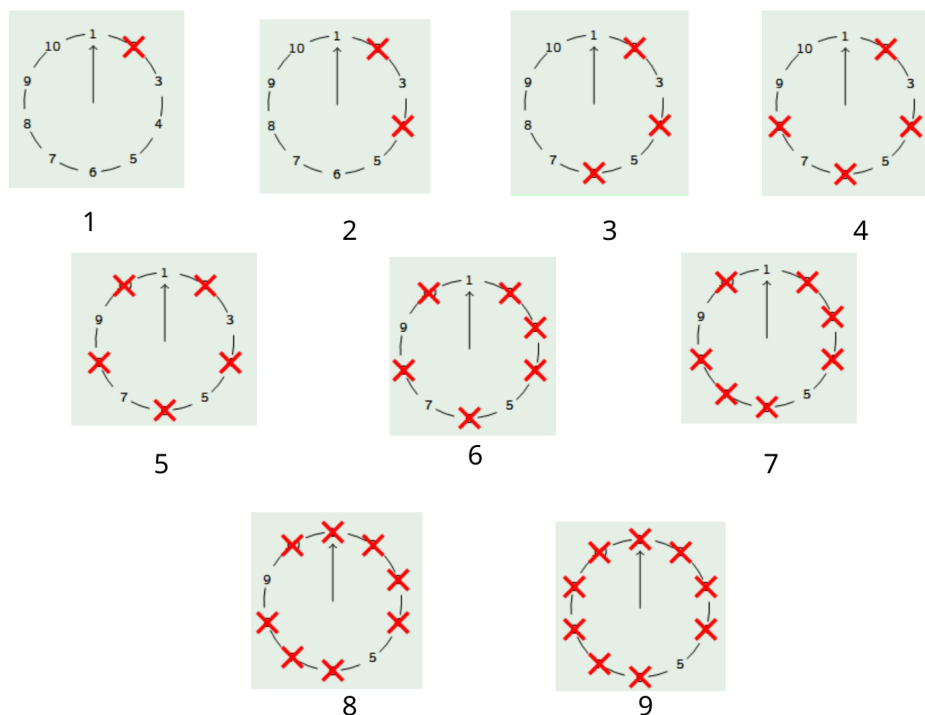


Figure 2: *

Lecture slide visual: $n = 10$ Josephus elimination. Red crosses show elimination step-by-step; position 5 survives.

How to explain in exam: People 1 to 10 are arranged clockwise. Starting from 1, every second person is removed. Counting continues around the remaining circle until one person remains.

Computer Science Uses:

- Circular data structures / circular linked list simulation.
- Process scheduling in circular order.
- Token-ring style network simulation.
- Turn-based elimination or game logic.

4.2 Closed Formula and CT: Find $J(196)$

*** MUST DO

Formula / Rule First

Write

$$n = 2^m + l, \quad 0 \leq l < 2^m.$$

Then

$$\boxed{J(n) = 2l + 1}.$$

How to solve this type

Question-e $J(n)$ dile:

1. n -er cheye choto ba equal largest power of 2 ber korba.
2. $l = n - 2^m$.
3. $J(n) = 2l + 1$.

For 196:

$$128 = 2^7 \leq 196 < 256.$$

So,

$$l = 196 - 128 = 68.$$

Then

$$J(196) = 2(68) + 1 = 137.$$

$$\boxed{J(196) = 137}.$$

Same-Type Practice

Find $J(100)$ and $J(50)$.

Practice Solution

For 100:

$$64 = 2^6, \quad l = 100 - 64 = 36,$$

$$J(100) = 2(36) + 1 = \boxed{73}.$$

For 50:

$$32 = 2^5, \quad l = 18,$$

$$J(50) = 2(18) + 1 = \boxed{37}.$$

4.3 CT Proof: $J(2^m + l) = 2l + 1$ by Induction

*** MUST DO

Formula to prove

$$J(2^m + l) = 2l + 1, \quad 0 \leq l < 2^m.$$

How to solve this type

Ei proof-ta first time-e confusing lagte pare. Main idea holo n ke two cases-e bhanga: **even** and **odd**. Josephus-er recurrence use korbo:

$$J(2n) = 2J(n) - 1, \quad J(2n + 1) = 2J(n) + 1.$$

Base Case: $m = 0$. Tokhon $2^0 = 1$ and only possible $l = 0$.

$$J(1) = 1.$$

Formula-o dey

$$2(0) + 1 = 1.$$

So base case true.

Induction Hypothesis: Dhoru formula m -er jonno true, i.e.

$$J(2^m + l) = 2l + 1, \quad 0 \leq l < 2^m.$$

Induction Step: Ekhon next block 2^{m+1} theke $2^{m+2} - 1$ -er number-gular jonno recurrence apply kori. Any number-ke parity onujayi $2q$ ba $2q + 1$ lekha jay. Smaller problem q -er survivor induction hypothesis diye jana. Even case-e

$$J(2q) = 2J(q) - 1,$$

and odd case-e

$$J(2q + 1) = 2J(q) + 1.$$

Hypothesis substitute korle survivor always “power-of-two part bad diye baki offset-er double + 1” form-e ashe. Tai next block-er jonno-o

$$J(2^{m+1} + l) = 2l + 1$$

valid thake. Hence mathematical induction diye formula proved.

Exam shortcut: Proof-er structure mone rakhba: *base case* $\rightarrow$ *assume formula* $\rightarrow$ *even/odd recurrence use* $\rightarrow$ *same* $2l + 1$ *form*. Numerical question-e proof lagbe na; direct largest power of 2 method use korba.

Same-Type Practice

Formula use kore $J(64)$, $J(65)$, $J(127)$ ber koro.

Practice Solution

$$J(64) = 1 \quad (l = 0),$$

$$J(65) = 2(1) + 1 = 3,$$

$$J(127) : 127 = 64 + 63 \Rightarrow J = 2(63) + 1 = 127.$$

4.4 Even and Odd Recurrence

**** IMPORTANT**

Formula / Rule First

$$J(2n) = 2J(n) - 1$$

$$J(2n + 1) = 2J(n) + 1.$$

Banglish idea: Even total people hole first complete round-e shob even-numbered position remove hoy. Remaining odd positions ke smaller Josephus problem-er position-er sathe map kora jay. Ei mapping thekei recurrence ashe.

5. Summation

5.1 Summation Basics

***** MUST DO**

Definition (Easy English). Summation is a compact notation for adding a sequence of terms. The symbol Σ represents the sum.

Example:

$$\sum_{i=1}^5 i = 1 + 2 + 3 + 4 + 5 = 15.$$

Formula / Rule First

$$\boxed{\sum_{i=1}^n i = \frac{n(n+1)}{2}}$$

$$\sum_{i=1}^n 1 = n, \quad \sum_{i=1}^n c = cn.$$

Same-Type Practice

$$\sum_{i=1}^{10} i$$

Practice Solution

$$\frac{10(11)}{2} = \boxed{55}.$$

Same-Type Practice

$$\sum_{i=1}^5 3i$$

Practice Solution

$$3 \sum_{i=1}^5 i = 3 \left(\frac{5 \cdot 6}{2} \right) = 3(15) = \boxed{45}.$$

5.2 Important Summation Laws

**** IMPORTANT**

$$\begin{aligned} \sum (a_i + b_i) &= \sum a_i + \sum b_i \\ \sum ca_i &= c \sum a_i. \end{aligned}$$

Meaning: Sum-er vitore addition thakle separate sum kora jay. Constant factor thakle constant-ke sum-er baire ana jay.

5.3 Multiple Sums and Predicate

***** MUST DO**

Definition (Easy English). An indicator is 1 when a condition is true and 0 when the condition is false. A predicate specifies which index pairs are allowed in a sum.

How to solve this type

$P(j, k)$ condition thakle first valid (j, k) pair ber korba. Tarpore only valid pair-er term add/count korba. Row-wise ba column-wise korleo same valid pairs thakle same answer.

Same-Type Practice

$j, k \geq 1$ and $j + k \leq 4$. Valid ordered pairs koyta?

Practice Solution

$$(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (3, 1).$$

Total:

$$\boxed{6}.$$

5.4 Stirling, Eulerian and Harmonic Numbers

**** IMPORTANT**

Definition (Easy English). A Stirling number of the second kind, $S(n, k)$, counts the ways to partition n distinct objects into k non-empty unlabeled subsets.

Memory clue: Stirling = distinct objects ke non-empty group-e partition.

Definition (Easy English). An Eulerian number counts permutations of n elements having a specified number of ascents (equivalently, a related descent convention may be used).

Memory clue: Eulerian = permutation + ascent/descent count.

Definition (Easy English). The n -th harmonic number is the sum of the reciprocals of the first n positive integers.

$$H_n = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n} = \sum_{k=1}^n \frac{1}{k}.$$

Same-Type Practice Find H_4 .

Practice Solution

$$H_4 = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} = \frac{12 + 6 + 4 + 3}{12} = \boxed{\frac{25}{12}}.$$

6. Number Theory: Integer Division, GCD, LCM

6.1 Integer Division

*** MUST DO

Definition (Easy English). For an integer a and a positive integer n , there are unique integers q and r such that $a = qn + r$, where $0 \leq r < n$.

Formula / Rule First

$$a = qn + r, \quad 0 \leq r < n.$$

a = dividend, n = divisor, q = quotient, r = remainder.

Same-Type Practice 47 ke 6 diye divide kore q, r ber koro.

Practice Solution

$$47 = 7(6) + 5.$$

So,

$$\boxed{q = 7, \quad r = 5}.$$

6.2 Divisibility

** IMPORTANT

Definition (Easy English). For integers a and b , a divides b , written $a \mid b$, if there exists an integer k such that $b = ak$.

Example:

$$3 \mid 15$$

because

$$15 = 3(5).$$

6.3 Prime Factorization, GCD and LCM

*** MUST DO

Definition (Easy English). Prime factorization means expressing a positive integer as a product of prime numbers.

Formula / Rule First

For prime-factor method:

- **GCD:** common prime factors-er **lowest exponent**.
- **LCM:** all required prime factors-er **highest exponent**.

Example:

$$60 = 2^2 \cdot 3 \cdot 5, \quad 90 = 2 \cdot 3^2 \cdot 5.$$

So,

$$\gcd(60, 90) = 2^1 \cdot 3^1 \cdot 5 = 30.$$

$$\text{lcm}(60, 90) = 2^2 \cdot 3^2 \cdot 5 = 180.$$

Same-Type Practice Find GCD and LCM of 24 and 36.

Practice Solution

$$24 = 2^3 \cdot 3, \quad 36 = 2^2 \cdot 3^2.$$

$$\gcd(24, 36) = 2^2 \cdot 3 = \boxed{12}.$$

$$\text{lcm}(24, 36) = 2^3 \cdot 3^2 = \boxed{72}.$$

7. Euclidean and Extended Euclidean Algorithm

7.1 Euclidean Algorithm

*** MUST DO

Definition (Easy English). The Euclidean algorithm finds $\gcd(a, b)$ by repeatedly replacing the pair with the divisor and remainder until the remainder becomes zero.

Formula / Rule First

$$\boxed{\gcd(a, b) = \gcd(b, a \bmod b)}$$

and

$$\boxed{\gcd(a, 0) = a}.$$

How to solve this type

Boro number ke choto number diye divide. Remainder ber koro. Next line-e ager divisor first number, remainder second number. Remainder 0 na howa porjonto repeat. **Last non-zero remainder = GCD.**

Example:

$$48 = 2(18) + 12$$

$$18 = 1(12) + 6$$

$$12 = 2(6) + 0.$$

Last non-zero remainder 6.

$$\boxed{\gcd(48, 18) = 6}.$$

Same-Type Practice Find $\gcd(2740, 1760)$.

Practice Solution

$$2740 = 1(1760) + 980$$

$$1760 = 1(980) + 780$$

$$980 = 1(780) + 200$$

$$780 = 3(200) + 180$$

$$200 = 1(180) + 20$$

$$180 = 9(20) + 0.$$

Therefore,

$$\boxed{\gcd(2740, 1760) = 20}.$$

7.2 Extended Euclidean Algorithm - Sir's Tabular Method for s, t

*** MUST DO

Definition (Easy English). The Extended Euclidean Algorithm finds $\gcd(a, b)$ and, at the same time, integers s and t such that $\gcd(a, b) = as + bt$. The numbers s and t are called Bezout coefficients.

Formula / Rule First

$$r = r_1 \bmod r_2$$

$$q = \left\lfloor \frac{r_1}{r_2} \right\rfloor$$

$$\boxed{s = s_1 - qs_2}, \quad \boxed{t = t_1 - qt_2}.$$

Initial values:

$$\boxed{s_1 = 1, s_2 = 0}, \quad \boxed{t_1 = 0, t_2 = 1}.$$

How to solve this type

Eta sir-er slide-er **tabular method**. Back-substitution na kore table-er row fill kore directly s, t ber korba.

1. First row-e $r_1 = a, r_2 = b$. Initial $s_1 = 1, s_2 = 0, t_1 = 0, t_2 = 1$.
2. $q = \lfloor r_1/r_2 \rfloor$ and $r = r_1 \bmod r_2$ ber koro.
3. Same row-e $s = s_1 - qs_2$ and $t = t_1 - qt_2$.
4. Next row-er jonno shift: $r_1 \leftarrow r_2, r_2 \leftarrow r; s_1 \leftarrow s_2, s_2 \leftarrow s; t_1 \leftarrow t_2, t_2 \leftarrow t$.
5. Remainder $r = 0$ howa porjonto repeat.
6. Final bottom state-e r_1 holo GCD, s_1 holo final s , and t_1 holo final t .

Memory shortcut: Prottek row-e age q, r ; tarpor same q diye s and t calculate. Then values left-e shift kore next row.

Slide Example: $a = 161, b = 28$ - Full Table

Goal:

$$\gcd(161, 28) = 161s + 28t.$$

Row 1:

$$q = \left\lfloor \frac{161}{28} \right\rfloor = 5, \quad r = 161 \bmod 28 = 21.$$
$$s = 1 - 5(0) = 1, \quad t = 0 - 5(1) = -5.$$

Row 2: Shift previous values. Now $r_1 = 28, r_2 = 21, s_1 = 0, s_2 = 1, t_1 = 1, t_2 = -5$.

$$q = \left\lfloor \frac{28}{21} \right\rfloor = 1, \quad r = 28 \bmod 21 = 7.$$
$$s = 0 - 1(1) = -1, \quad t = 1 - 1(-5) = 6.$$

Row 3: Now $r_1 = 21, r_2 = 7, s_1 = 1, s_2 = -1, t_1 = -5, t_2 = 6$.

$$q = \left\lfloor \frac{21}{7} \right\rfloor = 3, \quad r = 0.$$
$$s = 1 - 3(-1) = 4, \quad t = -5 - 3(6) = -23.$$

Full table:

q	r_1	r_2	r	s_1	s_2	s	t_1	t_2	t
5	161	28	21	1	0	1	0	1	-5
1	28	21	7	0	1	-1	1	-5	6
3	21	7	0	1	-1	4	-5	6	-23
	7	0		-1	4		6	-23	

Last non-zero remainder:

$$\boxed{\gcd(161, 28) = 7}.$$

Final coefficients bottom row theke:

$$\boxed{s = -1, \quad t = 6}.$$

Check:

$$161(-1) + 28(6) = -161 + 168 = 7.$$

Exam Writing In an exam, if the question asks for s, t , draw the table with columns $q, r_1, r_2, r, s_1, s_2, s, t_1, t_2, t$. Initialize $(s_1, s_2) = (1, 0)$ and $(t_1, t_2) = (0, 1)$, then use $s = s_1 - qs_2$ and $t = t_1 - qt_2$ row by row.

Same-Type Practice 1: Find $\gcd(35, 12)$ and s, t by Table

Initialize:

$$r_1 = 35, r_2 = 12; \quad s_1 = 1, s_2 = 0; \quad t_1 = 0, t_2 = 1.$$

q	r_1	r_2	r	s_1	s_2	s	t_1	t_2	t
2	35	12	11	1	0	1	0	1	-2
1	12	11	1	0	1	-1	1	-2	3
11	11	1	0	1	-1	12	-2	3	-35
	1	0		-1	12		3	-35	

Therefore,

$$\boxed{\gcd(35, 12) = 1, \quad s = -1, \quad t = 3}.$$

Check:

$$35(-1) + 12(3) = 1.$$

Same-Type Practice 2: Find $\gcd(99, 78)$ and s, t

q	r_1	r_2	r	s_1	s_2	s	t_1	t_2	t
1	99	78	21	1	0	1	0	1	-1
3	78	21	15	0	1	-3	1	-1	4
1	21	15	6	1	-3	4	-1	4	-5
2	15	6	3	-3	4	-11	4	-5	14
2	6	3	0	4	-11	26	-5	14	-33
	3	0		-11	26		14	-33	

So,

$$\boxed{\gcd(99, 78) = 3, \quad s = -11, \quad t = 14}.$$

Check:

$$99(-11) + 78(14) = -1089 + 1092 = 3.$$

7.3 More GCD/LCM Practice

**** IMPORTANT**

Same-Type Practice Find GCD and LCM of 72 and 120 by prime factorization.

Practice Solution

$$72 = 2^3 \cdot 3^2, \quad 120 = 2^3 \cdot 3 \cdot 5.$$

Lowest common exponents:

$$\gcd(72, 120) = 2^3 \cdot 3 = \boxed{24}.$$

Highest exponents:

$$\text{lcm}(72, 120) = 2^3 \cdot 3^2 \cdot 5 = \boxed{360}.$$

Same-Type Practice Find $\gcd(414, 662)$ using Euclidean Algorithm.

Practice Solution

$$662 = 1(414) + 248$$

$$414 = 1(248) + 166$$

$$248 = 1(166) + 82$$

$$166 = 2(82) + 2$$

$$82 = 41(2) + 0.$$

Thus

$$\boxed{\gcd(414, 662) = 2}.$$

8. Modular Arithmetic

8.1 Modulo and Residue

***** MUST DO**

Definition (Easy English). $a \bmod n$ is the non-negative remainder obtained when integer a is divided by positive integer n .

Formula / Rule First

$$\boxed{a \bmod n = a - n \left\lfloor \frac{a}{n} \right\rfloor}$$

and standard remainder satisfies

$$\boxed{0 \leq a \bmod n < n}.$$

Beginner shortcut for negative number: Modulus n repeatedly add korte thako jotokkhon answer 0 theke $n - 1$ range-e ashe.

8.2 Negative Modulo

***** MUST DO**

Example:

$$-18 \bmod 14.$$

Add 14:

$$-18 + 14 = -4.$$

Still negative, so abar 14 add:

$$-4 + 14 = 10.$$

Now $0 \leq 10 < 14$.

$$\boxed{-18 \bmod 14 = 10}.$$

Same-Type Practice

$$-7 \bmod 10$$

Practice Solution

$$-7 + 10 = 3.$$

$$\boxed{-7 \bmod 10 = 3}.$$

Same-Type Practice

$$-25 \bmod 7$$

Practice Solution 28 add korle:

$$-25 + 28 = 3.$$

$$\boxed{-25 \bmod 7 = 3}.$$

8.3 Congruence

***** MUST DO**

Definition (Easy English). Two integers a and b are congruent modulo n if their difference is divisible by n .

Formula / Rule First

$$\boxed{a \equiv b \pmod{n} \iff n \mid (a - b)}.$$

Example:

$$-18 \equiv 10 \pmod{14}$$

because

$$10 - (-18) = 28$$

and $14 \mid 28$.

Same-Type Practice

Are 17 and 5 congruent modulo 6?

Practice Solution

$$17 - 5 = 12.$$

Since $6 \mid 12$,

$$\boxed{17 \equiv 5 \pmod{6}}.$$

8.4 Operations in $\mathbb{Z}_n$

**** IMPORTANT**

How to solve this type

Normal addition/subtraction/multiplication korba, tarpor final result-er mod n niba.

Same-Type Practice

$$7 + 14 \text{ in } \mathbb{Z}_{15}.$$

Practice Solution

$$21 \bmod 15 = \boxed{6}.$$

Same-Type Practice

$$7 - 11 \text{ in } \mathbb{Z}_{13}.$$

Practice Solution

$$-4 \bmod 13 = 9.$$

$$\boxed{9}.$$

Same-Type Practice

$$11 \cdot 7 \text{ in } \mathbb{Z}_{20}.$$

Practice Solution

$$77 \bmod 20 = \boxed{17}.$$

8.5 Method of Squares / Exponentiation by Squaring

*** MUST DO

Definition (Easy English). The method of squares computes a large power modulo n efficiently by repeatedly squaring and using the binary or power-of-two decomposition of the exponent.

How to solve this type

1. Exponent-ke powers of 2-er sum-e bhango.
2. $a^1, a^2, a^4, a^8, \dots$ modulo n calculate koro by repeated squaring.
3. Exponent-e je powers lagbe shudhu segula multiply koro.
4. Final product-er mod nao.

Example: Find

$$3^{13} \bmod 7.$$

Exponent:

$$13 = 8 + 4 + 1.$$

Now:

$$3^1 \bmod 7 = 3$$

$$3^2 \bmod 7 = 9 \bmod 7 = 2$$

$$3^4 \bmod 7 = 2^2 \bmod 7 = 4$$

$$3^8 \bmod 7 = 4^2 \bmod 7 = 16 \bmod 7 = 2.$$

Therefore,

$$\begin{aligned} 3^{13} \bmod 7 &= (3^8 \cdot 3^4 \cdot 3) \bmod 7 \\ &= (2 \cdot 4 \cdot 3) \bmod 7 = 24 \bmod 7 = \boxed{3}. \end{aligned}$$

Same-Type Practice Find $2^{10} \bmod 7$.

Practice Solution

$$10 = 8 + 2.$$

$$2^2 \bmod 7 = 4, \quad 2^4 \bmod 7 = 16 \bmod 7 = 2, \quad 2^8 \bmod 7 = 4.$$

So,

$$2^{10} \bmod 7 = (2^8 \cdot 2^2) \bmod 7 = (4 \cdot 4) \bmod 7 = 16 \bmod 7 = \boxed{2}.$$

9. Additive and Multiplicative Inverse

9.1 Additive Inverse

** IMPORTANT

Definition (Easy English). In $\mathbb{Z}_n$, b is the additive inverse of a if $a + b \equiv 0 \pmod{n}$.

Formula / Rule First

$$\boxed{a + b \equiv 0 \pmod{n}}.$$

Example: additive inverse of 3 in $\mathbb{Z}_{10}$:

$$3 + 7 = 10 \equiv 0 \pmod{10}.$$

So inverse is

$$\boxed{7}.$$

9.2 Multiplicative Inverse

*** MUST DO

Definition (Easy English). In $\mathbb{Z}_n$, b is the multiplicative inverse of a if $ab \equiv 1 \pmod{n}$.

Formula / Rule First

$$ab \equiv 1 \pmod{n}.$$

Most important existence condition:

$$\gcd(a, n) = 1.$$

How to solve this type

First $\gcd(a, n)$ check korba. GCD 1 na hole immediately “inverse does not exist.” GCD 1 hole small modulus-e trial kora jay; boro number-e Extended Euclidean Algorithm best.

Example: Does 8 have inverse in $\mathbb{Z}_{10}$?

$$\gcd(8, 10) = 2 \neq 1.$$

Therefore,

$$\text{No multiplicative inverse exists.}$$

Same-Type Practice Find inverse of 3 in $\mathbb{Z}_{10}$.

Practice Solution Need $3b \equiv 1 \pmod{10}$.

$$3(7) = 21 \equiv 1 \pmod{10}.$$

So,

$$3^{-1} \equiv 7 \pmod{10}.$$

9.3 Multiplicative Inverse using the Same s, t Table

*** MUST DO

Formula / Rule First Inverse of a modulo n exists iff

$$\gcd(a, n) = 1.$$

Extended Euclidean table gives

$$1 = as + nt.$$

Therefore,

$$a^{-1} \equiv s \pmod{n}.$$

How to solve this type

Exactly ager q, r, s, t table use korba. Just input order hobe $r_1 = n$ and $r_2 = a$ if you want the table to follow the slide-style division. Final Bezout coefficient jeita a -er sathe multiply hoy, sheitai inverse; negative hole mod n kore positive residue niba.

Example: Inverse of 11 modulo 26 - Tabular Method

Need:

$$11x \equiv 1 \pmod{26}.$$

First,

$$\gcd(11, 26) = 1,$$

so inverse exists.

Use $a = 26, b = 11$ in the table. Then final identity will be $1 = 26s + 11t$, so t is the inverse coefficient of 11.

q	r_1	r_2	r	s_1	s_2	s	t_1	t_2	t
2	26	11	4	1	0	1	0	1	-2
2	11	4	3	0	1	-2	1	-2	5
1	4	3	1	1	-2	3	-2	5	-7
3	3	1	0	-2	3	-11	5	-7	26
	1	0		3	-11		-7	26	

Bottom row gives

$$1 = 26(3) + 11(-7).$$

So coefficient of 11 is -7 :

$$11^{-1} \equiv -7 \pmod{26}.$$

Convert to standard positive residue:

$$-7 \bmod 26 = 19.$$

Hence,

$$\boxed{11^{-1} \equiv 19 \pmod{26}}.$$

Check:

$$11(19) = 209 = 8(26) + 1.$$

Same-Type Practice Find inverse of 7 modulo 20 using the table.

Practice Solution Use $r_1 = 20, r_2 = 7$.

q	r_1	r_2	r	s_1	s_2	s	t_1	t_2	t
2	20	7	6	1	0	1	0	1	-2
1	7	6	1	0	1	-1	1	-2	3
6	6	1	0	1	-1	7	-2	3	-20
	1	0		-1	7		3	-20	

Thus

$$1 = 20(-1) + 7(3),$$

so

$$\boxed{7^{-1} \equiv 3 \pmod{20}}.$$

9.4 Slide-Type Example: Inverse of 23 modulo 100 - Tabular Method *** MUST DO

Formula / Rule First

$$\gcd(23, 100) = 1 \Rightarrow \text{inverse exists.}$$

Use $r_1 = 100, r_2 = 23$. Final identity $1 = 100s + 23t$ hole t -i 23-er inverse coefficient.

q	r_1	r_2	r	s_1	s_2	s	t_1	t_2	t
4	100	23	8	1	0	1	0	1	-4
2	23	8	7	0	1	-2	1	-4	9
1	8	7	1	1	-2	3	-4	9	-13
7	7	1	0	-2	3	-23	9	-13	100
	1	0		3	-23		-13	100	

Bottom row:

$$1 = 100(3) + 23(-13).$$

So

$$23^{-1} \equiv -13 \pmod{100}.$$

Positive residue:

$$-13 \bmod 100 = 87.$$

Therefore,

$$\boxed{23^{-1} \equiv 87 \pmod{100}}.$$

Check:

$$23(87) = 2001 = 20(100) + 1.$$

Exam cue: “Find multiplicative inverse using Extended Euclidean Algorithm” dekhlei ei q, r, s, t table draw korba.

9.5 More Modular Practice

**** IMPORTANT**

Same-Type Practice Find $-38 \bmod 11$.

Practice Solution Add a multiple of 11 until standard range 0 to 10:

$$-38 + 44 = 6.$$

So

$$\boxed{-38 \bmod 11 = 6}.$$

Same-Type Practice Check whether $41 \equiv 5 \pmod{12}$.

Practice Solution

$$41 - 5 = 36,$$

and $12 \mid 36$. Therefore

$$\boxed{41 \equiv 5 \pmod{12}}.$$

Same-Type Practice Find $5^{13} \bmod 13$ using repeated squaring.

Practice Solution

$$13 = 8 + 4 + 1.$$

$$5^2 \equiv 25 \equiv 12 \pmod{13},$$

$$5^4 \equiv 12^2 = 144 \equiv 1 \pmod{13},$$

$$5^8 \equiv 1^2 \equiv 1 \pmod{13}.$$

So

$$5^{13} \equiv 5^8 \cdot 5^4 \cdot 5 \equiv 1 \cdot 1 \cdot 5 = \boxed{5} \pmod{13}.$$

10. Mixed Exam Practice

Age answer na dekhe try korba. Question dekhei formula/method identify korar habit build korba.

1. Solve $T(n) = T(n/4) + 2$, $T(1) = 3$ and find complexity.
2. Lines in Plane: Find $R(10)$.
3. Josephus: Find $J(75)$.
4. Tower of Hanoi: Minimum moves for $n = 7$.
5. Find $\gcd(252, 105)$ using Euclidean Algorithm.
6. Find $-43 \bmod 12$.
7. Is $29 \equiv 5 \pmod{8}$?
8. Does 12 have a multiplicative inverse modulo 35?
9. Find the inverse of 12 modulo 35.
10. Write H_5 in expanded form.

Answers with Short Working

1.

$$T(n) = T(n/4^k) + 2k.$$

$$n/4^k = 1 \Rightarrow k = \log_4 n.$$

$$\boxed{T(n) = 3 + 2 \log_4 n = \Theta(\log n)}.$$

2.

$$R(10) = \frac{10(11)}{2} + 1 = \boxed{56}.$$

3. Largest power of 2 below 75 is 64.

$$l = 75 - 64 = 11.$$

$$J(75) = 2(11) + 1 = \boxed{23}.$$

4.

$$T(7) = 2^7 - 1 = \boxed{127}.$$

5.

$$252 = 2(105) + 42$$

$$105 = 2(42) + 21$$

$$42 = 2(21) + 0.$$

$$\boxed{\gcd(252, 105) = 21}.$$

6.

$$-43 + 48 = 5.$$

$$\boxed{-43 \bmod 12 = 5}.$$

7.

$$29 - 5 = 24, \quad 8 \mid 24.$$

So yes:

$$\boxed{29 \equiv 5 \pmod{8}}.$$

8.

$$\gcd(12, 35) = 1.$$

So inverse exists.

9.

$$35 = 2(12) + 11, \quad 12 = 1(11) + 1.$$

$$1 = 12 - 11 = 12 - [35 - 2(12)] = 3(12) - 35.$$

$$\boxed{12^{-1} \equiv 3 \pmod{35}}.$$

10.

$$\boxed{H_5 = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5}}.$$

11. Final Self-Test: Beginner to Exam Ready

Nicher question-gula kono solution na dekhe korte parle core math preparation bhalo level-e chole ashbe.

1. Recurrence relation-er easy English definition lekho and base case explain koro.
2. $T(n) = T(n/5) + 6$, $T(1) = 4$ full substitution diye solve koro.
3. $T(n) = T(n-1) + 3n$, $T(0) = 0$ solve kore complexity bolo.
4. Tower of Hanoi-r 3 rules lekho, recurrence derive koro, $n = 6$ moves ber koro.
5. Lines in Plane-r assumptions lekho and $R(7)$ ber koro.
6. Josephus $J(150)$ ber koro.
7. $\sum_{i=1}^{20} i$ and $\sum_{i=1}^8 4i$ ber koro.
8. Stirling, Eulerian, Harmonic Number-er easy English definition lekho.
9. Euclidean Algorithm diye $\gcd(391, 299)$ ber koro.
10. Bezout form-e $\gcd(35, 12)$ lekho.
11. $-71 \pmod{15}$ ber koro.
12. $53 \equiv 8 \pmod{15}$ true kina prove koro.
13. $3^{20} \pmod{7}$ repeated squaring diye koro.
14. 18-er inverse modulo 35 exist kore kina bolo; korle inverse ber koro.

Quick answers:

- 2) $4 + 6 \log_5 n$, 3) $\frac{3n(n+1)}{2} = \Theta(n^2)$,
4) 63, 5) 29, 6) 45,
7) 210 and 144, 9) 23,
11) 4, 12) true since $45 = 3(15)$,
13) 2, 14) $2 (18 \cdot 2 = 36 \equiv 1 \pmod{35})$.

12. Last-Minute Formula and Definition Sheet

Must Memorize Formulas

$$\begin{aligned}T(n) &= T(n-1) + c \Rightarrow O(n) \\T(n) &= T(n/b) + c \Rightarrow \Theta(\log n) \\ \text{Tower: } T(n) &= 2T(n-1) + 1, \quad T(n) = 2^n - 1 \\ \text{Lines: } R(n) &= R(n-1) + n, \quad R(n) = \frac{n(n+1)}{2} + 1 \\ \text{Josephus: } n &= 2^m + l \Rightarrow J(n) = 2l + 1 \\ \sum_{i=1}^n i &= \frac{n(n+1)}{2} \\ a &= qn + r, \quad 0 \leq r < n \\ \gcd(a, b) &= \gcd(b, a \bmod b) \\ \gcd(a, b) &= as + bt \\ a &\equiv b \pmod{n} \iff n \mid (a - b) \\ a^{-1} \pmod{n} &\text{ exists iff } \gcd(a, n) = 1.\end{aligned}$$

Must Memorize Easy Definitions

Recurrence Relation: An equation that expresses a problem or running time in terms of smaller instances of the same problem.

Tower of Hanoi: A three-peg disk puzzle where all disks are moved under size and one-disk-at-a-time rules.

Lines in the Plane: The problem of finding the maximum regions formed by straight lines under the no-parallel and no-three-concurrent assumptions.

Josephus Problem: A circular elimination problem where every second person is repeatedly removed until one survivor remains.

Summation: A compact notation for adding a sequence of terms.

Stirling Number: $S(n, k)$ counts partitions of n distinct objects into k non-empty unlabeled subsets.

Eulerian Number: Counts permutations with a specified number of ascents (or corresponding descents).

Harmonic Number: H_n is the sum of reciprocals $1 + \frac{1}{2} + \dots + \frac{1}{n}$.

Divisibility: $a \mid b$ if $b = ak$ for some integer k .

Prime Factorization: Expressing a positive integer as a product of prime numbers.

Euclidean Algorithm: A repeated-remainder method for finding the GCD.

Bezout Identity: $\gcd(a, b) = as + bt$ for some integers s, t .

Modulo: The standard non-negative remainder after integer division.

Congruence: a and b are congruent modulo n when n divides $a - b$.

Multiplicative Inverse: b is the inverse of a modulo n when $ab \equiv 1 \pmod{n}$.

Exam-er age order: CT math $\rightarrow$ Recurrence $\rightarrow$ Josephus/Tower/Lines $\rightarrow$ Euclidean/EEA
 $\rightarrow$ Modulo/Inverse $\rightarrow$ Summation definitions.